

Quiz 4

Friday, September 25 — angle, nearest neighbors, and clustering

Last updated: September 20, 2026.

Related lectures: [Angle](#), [Neighbors](#), [Clustering](#) (Week 4). Same day: [Basis](#) (not on this quiz).

Related lab: [Lab 2](#) (Monday, September 28; builds on [Lab 1](#)).

Textbooks: [Boyd & Vandenberghe \(VMLS\)](#) · [Lachniet Octave PDF](#) · also on [Resources](#).

[Schedule](#)

Core

- B 3.4–style angle / cosine similarity (from Lecture 08)
- B 3.24; nearest-neighbor choice of distance vs angle (Lecture 09)
- B 4.1, 4.2, 4.3
- C4.1: Compute one assignment or update step of k -means on a tiny dataset.
- C4.2: Explain whether a nearest-neighbor or cluster result changes under feature rescaling (Labs 1–2 / Lectures 09–10).

Stretch

B 4.4; a second k -means iteration or a brief interpretation of a centroid.

What a quiz item might change

Three short feature vectors, which similarity measure is used, one feature's units, or the initial cluster representatives.

Note

Dependence, span, and basis (Lectures 12–14) appear on [Midterm 1](#) and again on [Quiz 5](#) (Fri Oct 9) — not required for this quiz.